Inventory Stockout Response SOP

SYN-SOP-04 | Version 1.0 | Status: Synthetic MVP guidance

Synthetic operational knowledge created for the Option C Logistics project demonstration. This document is not a real company policy and must not be used for live operational decisions.

Purpose	Guide escalation when a delayed inbound shipment may create inventory shortage or service risk.
Applies to	Supply-chain disruption agent demonstration, operations review, and RAG retrieval testing.
Decision authority	Human operations reviewer. The AI system provides evidence and recommendations only.
Evidence rule	Use only observed tool evidence, model output, and retrieved guidance. Do not invent operational facts.

1. Required inputs

- Current usable inventory and committed demand.
- Average or planned daily demand for the affected item or category.
- Expected inbound quantity and estimated disruption duration.
- Alternative supplier or route lead time, when available.
- Inventory data timestamp and source.

2. Inventory coverage

Inventory coverage days should be calculated from verified stock and demand data. If the data is unavailable, the agent must return inventory_coverage_days = null and state the limitation.

3. Response levels

No verified coverage	Obtain current inventory data before making a stockout conclusion.
More than 7 days	Monitor the shipment and review coverage if the expected delay changes.
4 to 7 days	Assess demand prioritization and alternative sourcing options.
1 to 3 days	Escalate potential stockout exposure and prepare mitigation actions.
0 days	Treat as critical service risk and require immediate human review.

4. Operational workflow

1. Validate the inventory snapshot and demand period.
2. Compare coverage days with the expected disruption period.

3. Identify customer orders or service commitments that may be affected.
4. Assess alternative supplier, transfer, substitution, or rerouting options.
5. Record the recommendation, trade-offs, owner, and review time.

5. Guardrails

- Do not invent inventory coverage when no source is connected.
- Do not recommend allocation changes without identifying the affected demand.
- Do not use historical delay probability as a substitute for an expected delay duration.
- Require human approval for stock reallocation, substitution, or customer-priority changes.